Comparative Study of Bias Detection Datasets

Text Mining — Course paper

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Abstract

Media bias detection is difficult because the perception of bias is highly subjective, and different annotators often rely on different linguistic cues when judging whether a sentence is biased. Although both crowd annotated and expert annotated data are commonly used in media bias research, the linguistic differences between these annotation sources are often treated as noise rather than being studied directly.

In this paper, we present a comparative study of expert annotated and crowd annotated bias datasets to investigate how biased language is characterized under different annotation schemes. We focus on sentences labeled as biased and extract four interpretable linguistic features: subjectivity, emotional intensity, hedging, and sentence length. Using these features, we apply unsupervised analysis methods, including PCA, UMAP, and kernel density estimation, together with supervised classification models such as Logistic Regression and Random Forest, to examine differences between the two annotation sources.

Our results show clear linguistic differences between expert and crowd annotations. The two groups occupy different regions in the feature space, and supervised models are able to distinguish expert labeled sentences from crowd labeled sentences with moderate accuracy. Sentence length emerges as the most influential feature, with expert annotations associated with longer and more structurally complex sentences containing subtle subjectivity, whereas crowd annotations are more strongly linked to emotional intensity and sensational language.

These findings suggest that annotation source has a significant impact on the linguistic patterns captured in bias datasets and should be carefully considered when interpreting results in media bias research.

Keywords

text mining, bias detection, Natural Language Processing,Linguistic Features

ACM Reference Format:
Anh Trinh and Nurifeiya Anwaier. 2025. Comparative Study of Bias Detection Datasets: Text Mining — Course paper. In *Proceedings of Text Mining Course 2025 (TM '25).* ACM, New York, NY, USA, 7 pages.

1 Introduction

In an era where information travels effortlessly across the digital landscape, the automatic detection of media bias has emerged as a defining challenge for Natural Language Processing (NLP). Bias in news articles influences public opinions, polarizes communities, marginalizes minorities, and erodes trust in journalism[1]. Therefore, developing systemic methods to identify biased language is not only an academic exercise, but a necessary application to the real-word. However, the inherent subjectivity of language complicates this task, since what one reader perceives as fact, another may view as prejudice.

The strategies used to establish ground truth labels for training models have evolved from simple consensus-based approaches[2] to rigorous expert definitions[3]. Lim initially addressed the data scarcity problem by introducing a crowd-sourced dataset[2] that relies on layperson consensus. Subsequently, Spinde countered this by introducing an expert-annotated dataset[3], arguing for the necessity of rigorous training to identify nuance. While both approaches rely on inter-rater agreement as quality proxy, Spinde recently systematized the theoretical landscape with a comprehensive taxonomy to clarify that bias can manifest through subtle linguistic mechanisms such as hedging and framing[1]. While these works provide the much needed data and definition, they also highlight a significant discrepancy: experts and laypeople frequently disagree on what constitutes bias. Although this divergence is well-documented in recent literature[4], computational approaches frequently treat these disagreements as noise to be resolved through aggregation or majority voting[4]. Therefore, there remains a need to explicitly model and interpret the linguistic characteristics that drive these distinct definitions of bias, rather than obscuring them.

In this work, we formulate media bias detection as a supervised text analysis task, focusing on understanding how different annotation sources influence model behavior and linguistic feature representations.We address this problem by training supervised machine learning models on expert- and crowd-annotated data, and by analyzing their decisions using engineered linguistic features and interpretable methods.

In this paper, we address this need by investigating the specific linguistic drivers that differentiate expert and crowd perceptions of media bias. We move beyond classification accuracy to perform an interpretable analysis using engineered features, specifically subjectivity, emotion, hedging, and complexity. By using explainable models (Logistics Regression and Random Forest) alongside with statistical visualization (UMAP, PCA and KDE), we aim to uncover the fingerprints in decision-making process of different annotators.

To evaluate the importance of linguistic features in distinguishing between expert and crowd perspectives, we conduct our experiment to address two primary research questions: First, what underlying linguistic characteristics (e.g., subjectivity, emotion, hedging, complexity) that structurally differentiate sentences annotated by the crowd from those annotated by experts? Second, can machine learning models, when trained explicitly on these linguistic features, able to effectively distinguish between expert-labeled and crowd-labeled bias?

The remainder of this paper is organized as follows. Section 2 reviews the related work on media bias. Section 3 describes the datasets and the preprocessing steps. Section 4 outlines our methodology, including feature engineering and model specifications. Section 5 presents the experimental results. Section 6 discusses the implications of our findings. Finally, Section 7 concludes the paper and suggests directions for future research.

2 Related work

Media bias detection has been historically constrained by two primary challenges: the scarcity of fine-grained, sentence-level datasets and the subjectivity inherent in defining bias[2]. Recent literature has addressed these issues through evolving annotation strategies, ranging from crowdsourcing[2] to expert labeling[3], along with the development of rigorous taxonomies[1].

2.1 Crowdsourcing and Perceived Bias

In early research on media bias detection, crowdsourcing was widely used to solve the problem of limited labeled data. Since bias is subjective, asking ordinary readers to annotate sentences was an efficient way to collect a large amount of data. One example is the BASIL dataset proposed by Lim et al. (2020)[2], which contains 966 news sentences annotated by crowd workers based on whether they perceive bias.

Lim et al. show that crowdsourced annotations mainly reflect how bias is perceived by general readers, rather than how bias is strictly defined by experts. Their study finds that crowd judgments are strongly influenced by factors such as prior knowledge of the news event, personal political views, and opinions about the news outlet. Therefore, crowd annotations often represent individual interpretations instead of a clear and shared definition of biased language.

Another important finding is that agreement among crowd annotators is relatively low. This suggests that without clear guidelines or training, annotators may confuse biased language with personal disagreement with the content. However, this low agreement does not mean that crowd annotations are useless. Instead, it shows that different readers naturally understand and perceive bias in different ways.

Overall, crowdsourced annotations have both advantages and limitations. They allow researchers to collect data at a large scale and help capture how the public perceives media bias. At the same time, the variability in annotations creates difficulties for machine learning models, especially when crowd labels are treated as a single ground truth. For this reason, later studies have explored alternative annotation methods, such as expert-labeled datasets and more standardized annotation guidelines, which reduce ambiguity but usually result in smaller datasets.

2.2 Expert Annotation and Taxonomies

To reduce the noise caused by crowdsourcing, later studies began to use expert annotation. Spinde point out that crowd workers usually do not receive enough training to clearly distinguish between emotional or polarizing language and actual media bias [3]. To address this problem, they introduced the BABE dataset, which contains 3,700 sentences annotated by trained media researchers. Compared to crowdsourced data, expert annotation shows much higher agreement among annotators (Krippendorff's   0.40), suggesting that experts apply more consistent standards when labeling bias.

Using expert annotators helps reduce ambiguity in bias labels and leads to a more controlled definition of media bias. However, this approach also has limitations. Expert annotation is time-consuming and difficult to scale, which results in smaller datasets. Therefore, expert-labeled data represent a more standardized but also more limited view of media bias.

In addition to building expert-annotated datasets, Spinde also proposed a Media Bias Taxonomy to clarify how bias should be defined [1]. The taxonomy separates different types of bias, such as linguistic bias related to word choice and framing, and contextual bias related to selection and coverage.

However, Spinde also note that bias is still strongly influenced by social and contextual factors, even with clear definitions and expert guidelines [1].

2.3 Automated Detection and "The Black Box"

In recent years, research on media bias detection has increasingly focused on Transformer-based models [1]. Neural models such as BERT and RoBERTa, trained on expert-labeled data or using distant supervision, have been shown to perform much better than traditional feature-based approaches. However, although these models achieve strong classification results, they are often considered "black boxes" because their decision-making process is difficult to interpret.

Both Lim and Spinde point out that high accuracy alone is not sufficient for bias detection, and that it is important to understand why a model predicts a sentence as biased [2][3]. Motivated by this limitation, our work focuses on improving interpretability rather than relying solely on black-box performance.

Specifically, we combine statistical visualization with linguistic feature engineering to analyze differences between expert and crowd annotations. We use dimensionality reduction methods such as PCA and UMAP, as well as density estimation techniques like KDE, to visualize structural differences between the two groups. In addition, we apply classical machine learning models to these features to examine whether they can distinguish between expert-labeled and crowd-labeled data. Overall, our approach prioritizes linguistic interpretability over purely performance-driven black-box models.

3 Data

To analyze the linguistic divergence between crowd and expert perceptions of bias, we combined data from two distinct sources, each representing one of our target group.

3.1 Crowd Annotations

For the crowdsourcing perspective, we used the dataset introduced by Lim et al. This corpus consists of 46 news articles covering four major political and social events: the NFL player protests, the Facebook data privacy scandal, North Korea–US relations, and the suicide of lawmaker Dan Johnson[2]. These topics were selected to reflect a broad range of contemporary and politically sensitive issues, where perceptions of media bias are likely to vary across readers.

The articles were collected from a diverse set of news outlets spanning the political spectrum, including sources such as The Washington Post and the Daily Mail. This diversity was intended to capture different writing styles, ideological positions, and framing strategies, thereby providing a realistic setting for analyzing crowd-based perceptions of media bias.

In total, the dataset contains 966 sentences. Each sentence was independently annotated by crowd workers recruited through the Figure Eight platform, with 28 annotators participating in the labeling process. Workers were asked to assess the degree of media bias present in each sentence using a four-point ordinal scale ranging from *Neutral* to *Very Biased*. Following prior work, we binarized these labels for our classification task, grouping lower scores as non-biased and higher scores as biased. This binarization allows for a direct comparison with expert annotations and supports the use of standard supervised classification models.

Overall, this dataset represents a crowd-based perspective on media bias, reflecting how non-expert readers interpret bias cues across different topics, outlets, and writing styles.

3.1.1 Label Distribution. Table 1 reports the frequency of each bias label in the crowd-annotated dataset.

Bias label	Number of sentences
1	32
2	72
3	91
4	19

Table 1: Distribution of bias labels in the crowd-annotated dataset.

Figure 1 visualizes the same distribution.

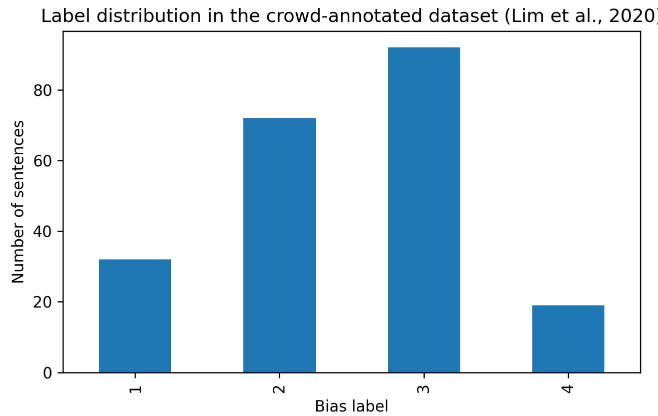

Figure 1: Distribution of bias labels in the crowd-annotated dataset.

3.2 Expert Annotations

For the expert perspective, we used the BABE (Bias Annotations by Experts) dataset introduced by Spinde. In contrast to the crowd data, this corpus was annotated by trained media researchers who followed a strict instruction guideline[3].

The BABE dataset is larger and more diverse in terms of topics, containing approximately 3,700 sentences covering 12 controversial topics from 14 major US news outlets. These annotations serve as our "Expert" class, representing a more standardized, rule-based definition of bias.

3.3 Example Sentences

To provide a concrete illustration of the data, we present example sentences from both the crowd annotated and expert annotated datasets. These examples are intended to show how media bias is expressed and labeled under different annotation schemes, rather than to support any experimental claims.

Table 2: Representative example sentences from the crowd and expert datasets

Source	Example sentence	Label
Crowd	"The controversial decision sparked outrage across the country."	Biased
Expert	"The policy was criticized by opposition parties for lacking transparency."	Biased

3.4 Preprocessing

To build an unified dataset for comparison, we applied the following preprocessign steps:

- **Text Cleaning:** The raw text in the crowd dataset contained indexing artifacts. We removed these using regular expressions to ensure that sentence length was not artificially inflated by metadata.
- **Aggregation and Binarization (Crowd):** Since the crowd data contained multiple ratings per sentence on a 4-point scale, we first aggregated these into a single mean score per sentence. To align this with binary classifications of the expert dataset, we applied a rule: we labeled a sentence as Biased (1) only if its mean rating exceeded 2.5. Sentences with a mean rating of 2.5 or lower were classified as Neutral

(0). This cutoff was selected to isolate sentences with strong signal of perceived bias.
- **Filtering (Expert):** For the expert dataset, we filtered the corpus to keep only sentences explicitly labeled as Biased.
- **Merging:** The two filtered subsets were concatenated into a single dataframe, with a new target variable Source created to distinguish between Crowd (Lim) and Expert (Spinde).

3.5 Limitations and Biases

While combining these datasets allows for a comparative analysis, we acknowledge that there are inherent biases and limitations in the data:

- **Topic Disparity:** The two datasets do not cover identical news events. The crowd data is restricted to four specific events, whereas the expert data covers a broader range of 12 topics. This can lead to a spurious correlation, whereby models differentiate between groups using topic specific keywords instead of relying solely on linguistic markers of bias.
- **Annotator Subjectivity:** Although we treat the expert labels as gold standard, media bias detection remains inherently subjective. Lim noted that crowd workers' perceptions were significantly influenced by their prior knowledge of the events[2]. Similarly, while expert guidelines were rigorous, they reflect a specific academic definition of bias that may not align with general public perception.
- **Sample Size Imbalance:** The expert dataset (N = 3,700) is significantly larger than the crowd dataset (N = 966). Therefore, the smaller sample size of the crowd class limits the diversity of layperson bias examples available for analysis.

4 Methodology

To investigate the linguistic divergence between expert and crowd annotations, we designed a pipeline that moves from exploratory feature analysis to predictive modeling. Our goal is to determine if the two groups operate on structurally different definitions of bias.

4.1 Feature Engineering

We transformed every biased sentence into a vector of four interpretable linguistic features. Unlike high-dimensional embeddings (e.g., BERT), these features were selected to serve as transparent indicators for specific cognitive and stylistic markers.

- **Subjectivity:** We used the TextBlob library to extract a subjectivity score between 0 and 1. This measures the degree to which a sentence expresses personal opinion rather than factual information.
- **Emotion (Absolute Polarity):** We calculated the sentiment polarity of the sentence and took its absolute value. This feature captures emotional intensity, treating highly positive and highly negative sentences as equally emotional.
- **Hedge Words:** Drawing from the linguistic bias taxonomy[1], we engineered a custom list of epistemological markers commonly used to soften claims. We computed the raw count of the following terms per sentence": *"maybe", "perhaps", "likely", "unlikely", "allegedly", "claimed", "apparently", "suggest", "seem", "possibly"*.
- **Complexity (Sentence Length):** We calculated the total word count per sentence. This structural feature was included to test the hypothesis that experts may prefer more complex formulations of bias, while the crowd may react to shorter and punchier statements.

4.2 Unsupervised Analysis

Before modeling, we visualized the manifold structure of the data to see if expert and crowd sentences occupy distinct regions of the feature space.

- **UMAP:** We used UMAP (Uniform Manifold Approximation and Projection) to project the 4-dimensional feature space into a 2D map ($n\_neighbors = 30, min\_dist = 0.1$). This technique preserves local neighborhood structures, allowing us to see if sentences cluster by annotator source.
- **PCA:** To interpret the global variance, we used PCA (Principle Component Analysis) on the features. We constructed a biplot to visualize the feature loadings, which reveals the correlation between specific features and the principal components that separate the groups.
- **Distributional Analysis:** We used KDE (Kernel Density Estimation) to plot the probability density functions of each feature for both groups. This allows for a direct visual comparison of how the average expert biased sentence differs from the average crowd biased sentence.

4.3 Supervised Classification

To quantify the divergence of the two groups, we formulated a binary classification task where the model predicts the annotation source ($0 = \text{Crowd}, 1 = \text{Expert}$) based only on the four linguistic features. We used two complementary models:

- **Random Forest Classifier:** We trained a Random Forest ($n\_estimators = 100$) to capture the non-linear interactions between features. We used Mean Decrease Impurity scores to rank which linguistic features were most critical for distinguishing between the two groups.
- **Logistic Regression:** We trained a linear classifier on standardized data to extract the regression coefficients. This provides directionality to our findings: a positive coefficient indicates a feature is associated with Experts, while a negative coefficient indicates an association with the Crowd.

Models were evaluated on a held-out test set (20% split) using accuracy as the primary metric.

5 Result

Our experiments reveal a significant structural divergence between crowd and expert annotations. The results show that even though features play a role, simple structural artifacts, namely sentence length, act as a dominant discriminator between the two groups.

5.1 Unsupervised Analysis Results

The unsupervised visualization of the feature space provides the first evidence of separation.

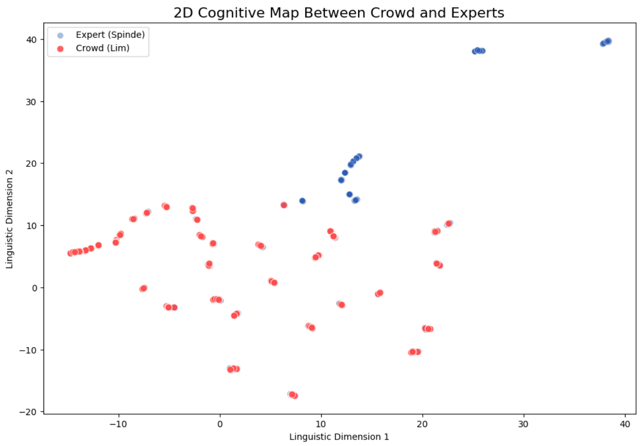

Figure 2: 2D Cognitive Map of Annotator Perspectives

UMAP Projection: Figure 2 displays the UMAP projection of all biased sentences. We observe two distinct, although overlapping, clusters. The expert sentences tend to occupy a denser area of the manifold, suggesting a more consistent or standardized definition of bias. On the other hand, the crowd sentences are more dispersed, reflecting the higher variance and noise inherent in crowd perception.

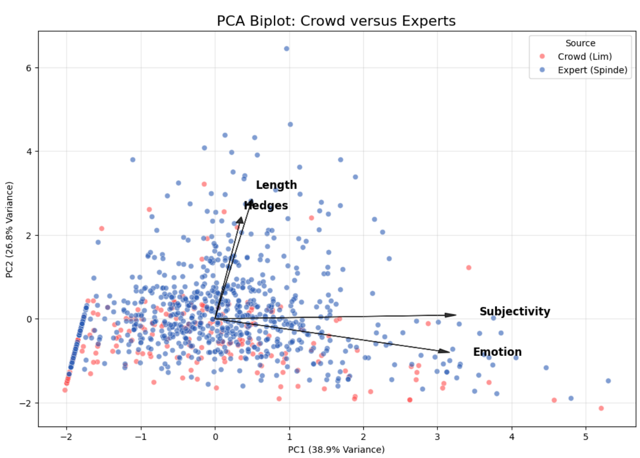

Figure 3: Principle Component Analysis Biplot

PCA and Feature Loadings: The PCA biplot (Figure 3) further highlights the drivers behind this separation. The first principal component (PC1) explains the majority of the variance and is strongly aligned with the Sentence Length vector. The biplot shows a clear pull: Expert sentences are pulled towards higher values of Length and Hedge, while crowd sentences are distributed towards emotion and subjectivity.

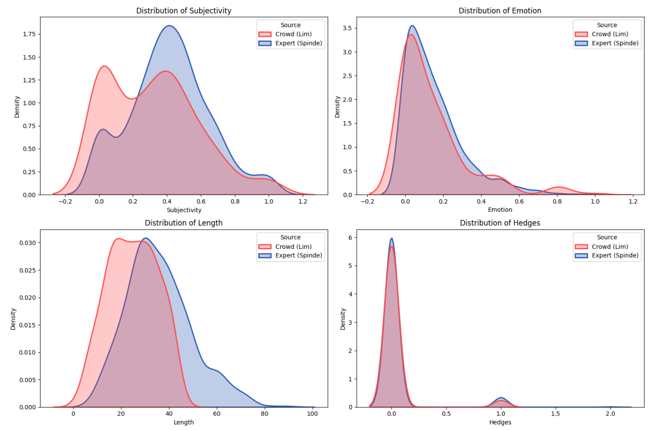

Figure 4: Distributional Divergence between Crowd and Expert

Distributional Divergence: The KDE plots reveal the specific nature of the artifact we discussed above. The most noticeable difference is observed in Sentence Length. The expert distribution is right-skewed with a significantly higher mean, while the crowd distribution is concentrated around shorter sentences. This suggests that experts are flagging complex sentences, while the crowd is reacting to shorter, "clickbait" sentences.

5.2 Classification Performance

To quantify this divergence, we trained our supervised models to predict the source of the annotation (0 = Crowd, 1 = Expert).
Model Accuracy: Despite the subjective nature of the task, the Random Forest classifier achieved an accuracy of **0.77** on the held-out test set. This confirms that the two groups are not just randomly different, but systematically distinct.

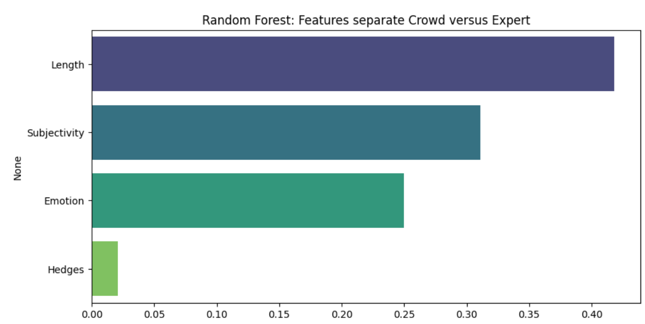

Figure 5: Random Forest Feature Importance (MDI)

Feature Importance (Random Forrest): Figure 5 displays the MDI scores. Consistent with our exploratory analysis, Sentence Length emerged as the single most predictive feature, significantly outperforming semantic features like Subjectivity or Emotion. This indicates that the model is "cheating" by using length as a rule to distinguish the groups.

Table 3: Logistic Regression coefficients. Positive values indicate Expert association, negative values indicate Crowd association

Feature	Coefficient
Length	1.025
Subjectivity	0.359
Hedges	-0.041
Emotion	-0.135

Directionality (Logistic Regression): As detailed in Table 3, the regression coefficients reveal distinct annotation strategies. We observe a strong postive association with Length (Coef: +1.025) and Subjectivity (Coef: 0.359), indicating that experts identify bias through both structural complexity and specific subjective framing. On the other hand, Emotion (Coef: -0.135) and Hedges (Coef: -0.041) yield negative coefficients, suggesting that the crowd perception is more driven by emotional markers.

6 Discussions

6.1 Structural Complexity and Annotator Perception

Our most significant finding is the dominance of sentence length as a discriminator between expert and crowd annotations. While this might be dismissed as a dataset artifact, it likely reflects a deeper cognitive divergence in how bias is perceived. Experts, who follow academic guidelines, appear to identify bias in complex, nuanced sentences. These often appear as longer and structurally elaborate sentences. In contrast, the crowd appears to react to shorter and punchier statements, potentially equating bias with the headline-centric style of sensationalist media.

This observation has critical implications for the black box models discussed in Section 2. If a simple feature like length can distinguish these groups with high accuracy, there is a risk that complex neural models are similarly using sentence complexity as a discriminator for bias, rather than learning semantic characteristics of bias itself.

6.2 Subjectivity versus Emotion

Our analysis highlights a nuance in how bias is flagged. Contrary to the expectation that experts are more objective, the positive coefficient for Subjectivity (0.359) indicates that experts are actually highly sensitive to subjective phrasing, likely identifying subtle opinionated framing within complex sentences. On the other hand, the crowd's association is defined by Emotion (-0.135) and Hedges (-0.041). This suggests a clear split in focus: the crowd detects inflammatory tone, while experts identify journalistic bias. Future research should arguably treat these not as conflicting ground truths, but as separate detection tasks.

6.3 Limitations and Future Work

Our primary limitation is the difference in subject matter between the two datasets. The expert data covers broad controversial topics, while the crowd data focuses on specific breaking events like scandals. It is possible that policy discussions naturally require longer sentences than event reporting. As a consequent, the sentence length difference we observed might simply reflect the topic rather than the annotator's preference. To fix this, future research must apply both annotation methods to the exact same set of articles. We also recommend that future benchmarks explicitly control the length of sentence to make sure that models are detecting bias and not just counting words.

7 Conclusion

This study provides a detailed comparison of how expert and crowd-sourced annotations differ in the task of media bias detection. Instead of treating annotation disagreement as noise that should be averaged out, we considered annotation source as a key variable and examined whether these differences can be identified using simple and interpretable linguistic features.

By combining unsupervised visualization methods such as UMAP and PCA with supervised machine learning models including Random Forest and Logistic Regression, we showed that expert- and crowd-annotated sentences are not only different at the label level, but also follow different linguistic patterns. Across all analyses, the two annotation sources occupy different regions in the feature space and can be separated with moderate accuracy using only four low-dimensional features.

Our results indicate that this separation is mainly driven by structural and stylistic aspects of language rather than deep semantic meaning. Sentence length was found to be the strongest factor distinguishing the two datasets. Experts are more likely to identify bias in longer, structurally complex, and more academically written sentences, while crowd annotators tend to focus on shorter, more direct, and emotionally charged statements, which often resemble language. The regression results further show that expert annotations are positively related to sentence length and subtle forms of subjectivity within complex framing. In contrast, crowd annotations are more strongly influenced by emotional intensity and inflammatory language, often associating sensational expressions with bias.

The relatively high classification accuracy achieved by our models (0.77) points to an important methodological concern for automated bias detection systems. Although such performance may seem encouraging, it suggests that models may rely on simple structural cues such as sentence length instead of learning meaningful semantic representations of bias. This highlights a broader "black box" risk, where increasing model complexity does not necessarily prevent shortcut learning, but may instead reinforce dataset-specific patterns and lead to misleading interpretations of model performance.

From a broader research perspective, our findings question the assumption that expert and crowd-sourced annotations represent different approximations of a single ground truth. Instead, they reflect two distinct but internally consistent detection tasks: one based on professional journalistic standards and careful framing, and the other shaped by general public perception and emotional salience. Treating one annotation source as universally superior oversimplifies the subjective nature of media bias and overlooks important differences in how bias is perceived and labeled.

This study also has several limitations that suggest directions for future research. Most importantly, future work should apply multiple annotation schemes to the same set of texts in order to better separate annotator perception from topical effects. In addition, future benchmark datasets should explicitly control for structural features such as sentence length, so that models are evaluated based on bias-related semantics rather than dataset-specific artifacts. More generally, annotation source should be treated as an explicit experimental factor in media bias detection research rather than a neutral design choice.

In conclusion, this work shows that progress in media bias detection depends not only on developing stronger models, but also on understanding how bias is defined and interpreted through the annotation process. By highlighting clear and interpretable differences between expert and crowd perspectives, we argue for a more annotation-aware approach to dataset design, model evaluation, and the interpretation of results in future bias detection research.

References

[1] Timo Spinde, Smi Hinterreiter, Fabian Haak, Terry Ruas, Helge Giese, Norman Meuschke, and Bela Gipp. The media bias taxonomy: A systematic literature review on the forms and automated detection of media bias, 2024.

[2] Sora Lim, Adam Jatowt, Michael Färber, and Masatoshi Yoshikawa. Annotating and analyzing biased sentences in news articles using crowdsourcing. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 1478–1484, Marseille, France, May 2020. European Language Resources Association.

[3] Timo Spinde, Manuel Plank, Jan-David Krieger, Terry Ruas, Bela Gipp, and Akiko Aizawa. Neural media bias detection using distant supervision with babe - bias annotations by experts. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, page 1166–1177. Association for Computational Linguistics, 2021.

[4] Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. Dealing with disagreements: Looking beyond the majority vote in subjective annotations, 2021.